Tribhuvan University

Faculty of Humanities and Social Sciences

Bachelor of Arts in Computer Application

Mahendra Multiple Campus Nepalgunj

Preboard Examination

2025 (2082 BS)

Course Title: Probability and Statistics
Code No: CAST202
Semester: Third

Full Marks: 60
Pass Marks: 24
Time: 3 hours

Candidates are required to answer the questions in their own words as far as possible.

Attempt any six questions:

Group B

[6×5=30]

2. What do you mean by primary and secondary data? Write different sources of primary and secondary data.

3. The following table represents the marks of 100 students.

Marks	0-20	20-40	20-60	60-80	80-100
No. of students	14	-	27	-	15

If the median is 48, find missing frequencies.

4. Find quartile deviation of given frequency distribution.

Height	Below 5	5 – 10	10 - 15	15- 20	20 - 25
Frequency	5	7	8	6	2

5. Calculate correlation coefficient between X and Y

X	12	27	16	22	25	20
Y	18	20	28	21	25	15

6. Fit the Binomial distribution from the following data where $p = 0.5$

No. of heads	0	1	2	3	4
Frequency	28	62	46	20	4

7. What do you mean by Random and Non-random Sampling? Define systematic sampling method.

8. Determine Q_1, D_6 and P_{85} of given data.

Class	10-20	20-30	30-40	40-50	50-60	60-70	70-80
Frequency	10	11	27	35	29	11	10

Group C

Attempt any two questions: **[2×10=20]**

9. The following table shows that the number of motor registration and sales of motor tyres by a wholesale dealer in Katmandu for term of 5 years.

Year	2015	2016	2017	2018	2019
Motor Registration(000)	60	63	72	75	80
No. Of Tyres Sold (000)	125	110	130	135	150

i. Estimate the sales of tyres when expected motor registration in next year is 90,000.

ii. Also calculate coefficient of correlation and interpret the result.

10. The following table gives the monthly income of the workers in Kathmandu and pokhara.

Monthly Income (000)	Number of workers	
	Kathmandu	Pokhara
45	5	32
50	15	22
55	20	15
60	20	12
65	12	10
70	10	6
75	9	2

Find which city has greater inequality in the distribution of monthly wage?

11. Given a normal distribution with mean 50 and sd 5, find the probability of

a. $P(X < 65)$ b. $P(X > 53)$ c. $P(46 < P < 56)$
d. $P(X < 35)$ e. $P(X < 40 \text{ or } X > 65)$
f. 10 % of the values are less than the X, then what is the value of X?

Best of Luck

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Pre-Board Examination1
2025 (2082 BS)

Course Title: Web Technology
Code No: CACS205
Semester: III

Full Marks: 60
Pass Marks: 24
Time: 3 hours

Candidates are required to answer the questions in their own words as far as possible.

Group B

Attempt any SIX questions: **[6×5=30]**

2. Explain the structure of a form in HTML. Mention the different types of form inputs with examples.
3. Differentiate between 2-Tier, 3-Tier, and n-Tier architecture.
4. Differentiate between XML Schema (XSD) and DTD with suitable examples.
5. What is the role of a web server in a web application architecture?
6. Explain form handling on the server side with an example.
7. Explain the concept of namespaces in XML with examples.
8. Compare and contrast Inline, Internal, and External CSS with examples.

Group C

Attempt any ~~SIX~~ TWO questions: **[6×5=30]**

9. Write HTML code to create a table with merged cells and custom background colors.
10. Write and explain SQL statements for SELECT, INSERT, UPDATE, and DELETE operations.
11. Describe the difference between simple and complex types in XML Schema.

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Pre-Board Examination
2025 (2082 BS)

Course Title: OOP in Java
Code No: CACS204
Semester: Third

Full Marks: 60
Pass Marks: 24
Time: 3 hours

Candidates are required to answer the questions in their own words as far as possible.

Group B
Attempt any SIX questions: [6×5=30]

2. Define OOP. Explain any two Java features. [1 + 2 + 2]
3. a) Define the java primitive data types with examples. [2.5]
   b) Write the operators used in java with examples. [2.5]
4. Define Abstract Class. Differentiate between method overloading and method overriding with suitable program as an example. [1 + 4]
5. How Class is differ from Interface? Write any program to implement multiple inheritance in Java. [1 + 4]
6. Define user defined exceptions. Write a program that includes a try block and a catch clause which processes the arithmetic exception generated by division-by-zero error. [1 + 4]
7. What is super key in java? Explain with an example. [1 + 4]
8. What is constructor in java? Explain its type with a suitable example.[1+4]

Group C
Attempt any two questions: [2×10=20]

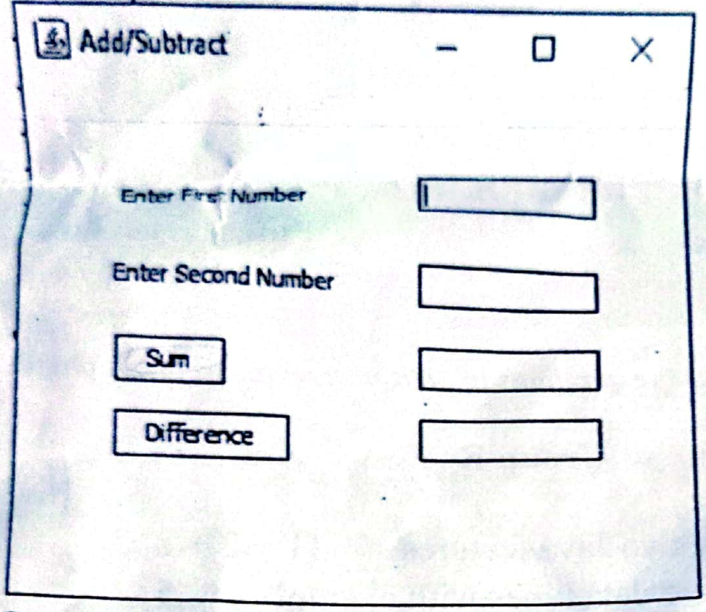
9. a) Define java thread. Explain java thread lifecycle. Explain with an example. [1 + 2 + 2]
   b) What is Applet ? Explain with an example. [1+ 4]

CamScanner

10. a) Define java package. Explain any one package with an example. [1+ 4]

b) Define Array. Explain the Array with an example. [1 + 4]

11. Write java code to design and function following frame using Java Swing.



Sum and Difference are two buttons they are used to calculate sum and difference and display the result in belonging text field. [5]

b) Explain the difference between awt and swing. [5]

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Pre-Board Examination
2025 (2082 BS)

Course Title: Data Structure and Algorithm
Code No: CACS201
Semester: Third

Full Marks: 60
Pass Marks: 24
Time: 3 hours

Candidates are required to answer the questions in their own words as far as possible.

Group B

Attempt any SIX questions: *[6×5=30]*

2. What is Data Structure? Show the status of stack converting following infix expression to post fix P + Q – (R*S/T+U)-V*W
3. Define circular queue? How does circular queue overcome the limitation of linear queue? Explain.
4. What is binary search? Consider a hash table of size 10; insert the keys 62, 37, 36, 44, 67, 91 and 107 using linear probing.
5. Define AVL tree. Construct AVL tree from given data set: 4, 6, 12, 9, 5, 2, 13, 8, 3, 7 ,11.
6. What is stack? List the applications of stack. Write an algorithm or procedure to perform PUSH and POP operation in stack.
7. What is recursion? Write an algorithm to solve Tower of Hanoi problem.
8. Draw a BST from the string DATASTRUCTURE and traverse the tree in post order and preorder.

CamScanner

Group C

Attempt any two questions: **[2×10=20]**

10. Explain quick sort algorithm with Big-oh notation in worst case and sort the given data using quick sort.

11. Data: 8, 10, 5, 12, 14, 5, 7, 13.

12. Explain the Dijkstra's algorithm with an example.

13. Define graph and tree data structure. Explain breadth first traversal and depth first traversal with example.

Best of Luck

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Pre-Board Examination
2024 (2080 BS)

Course Title: System Analysis and Design
Code No: CACS203
Semester: ~~Sixth~~ 3rd
Full Marks: 60
Pass Marks: 24
Time: 3 hours

Candidates are required to answer the questions in their own words as far as possible.

Group B

Attempt any Six questions. [6×5=30]

2. Compare and contrast between comparative analysis model and interactive service model.
3. Describe information systems and their types.
4. Describe corporate information system planning.
5. Explain normalization and its importance in database design.
6. Transform ER diagram to relations with examples.
7. Compare Agile and Waterfall. Why use Agile?
8. Explain the guidelines to design an interface and dialogue box for e-commerce system.
9. Write short notes on (any two):
   a. eXtreme Programming.
   b. system acquisition
   c. Gantt Chart

Group C

Attempt any two questions. [2×10=20]

10. What is process modeling? Draw DFD for an online hotel booking system up to level 2.
11. What qualities a software tester should have? What are the deliverables from coding, testing and installation? Explain different types of software testing in detail.
12. What is CASE? Explain its tools and their role in SDLC phases.